

Cocofest: an Open-Source Python Package for Functional Electrical Stimulation Optimization in Optimal Control

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DOI: 10.xxxxxx/draft

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Editor: Augusto Borges

Reviewers:

- @Aravind-Sundararajan
- @siboles

Submitted: 06 September 2025

Published: unpublished

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Summary

Functional electrical stimulation (FES) is a rehabilitation method intended to promote motor recovery notably after neurological impairment. Applying coordinated electrical pulses to muscles elicits functional movements like walking, reaching, and grasping. FES rehabilitation mostly relies on empirical settings, as responses to stimulation vary across populations and muscles. Empirical settings often cause overstimulation and premature fatigue (Ibitoye et al., 2016), shortening rehabilitation sessions and diminishing therapeutic benefit. Consequently, advanced control approaches like optimal control-driven FES are gaining interest in personalizing and improving FES rehabilitation efficiency, meanwhile delaying muscle fatigue (Co et al., 2025). To address this need, we designed Cocofest (Custom Optimal Control for Functional Electrical STimulation), an open-source Python package for optimal control-driven FES. Cocofest provides a framework to generate personalized pulse trains (Fig. 1) based on nonlinear dynamics models for FES, for several musculoskeletal models and motor tasks. The package includes over 10 examples, covering optimization of FES-related pulse train parameters (including frequency, pulse width, pulse intensity), FES model parameters identification from in-vivo measurements, and long duration predictive simulations.

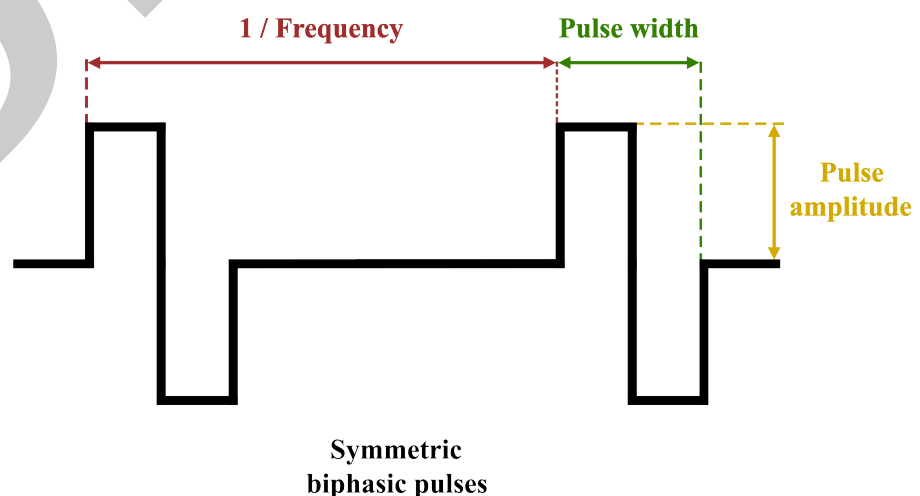


Figure 1: Pulse train parameters that can be optimized in Cocofest

Statement of Need

Since the pioneer study on optimal control-driven FES (Hunt et al., 1997), no code has been shared in the field, limiting objective comparison and replicability across studies. The lack of open-source practice led to an absence of consensus on how to choose nonlinear dynamics for FES, and which cost functions to use for dedicated clinical needs, hindering standardization and cumulative progress (Co et al., 2025). To address these challenges and support collective scientific progress, Cocofest fulfills the following four needs:

Firstly, the relationship between the pulse train parameters (e.g., frequency, pulse width and intensity; Fig. 1) and the resulting muscle force, joint torque, and muscle fatigue can be modeled with different nonlinear dynamics (Ding et al., 2003; Veltink et al., 1992). Gathering them within a unified package would facilitate comparison for more informed modelling choices.

Secondly, no study has compared different optimal control problem (OCP) formulations applied to FES, due to OCP implementation challenges (Co et al., 2025). Easily customizable OCP formulation, involving objective functions, models, and transcriptions is required to provide an adequate research framework. Having the possibility to switch between various OCP transcriptions (e.g., direct collocation or direct multiple shooting) is essential when dealing with stiff differential equations (Puchaud et al., 2023), often embedded in FES models. Muscle fatigue is the primary challenge in FES. Enabling the development and comparison of different OCP formulations could help address research questions, yield novel stimulation patterns and enhance fatigue reduction. Moreover, using receding-horizon estimation for longer simulations reduces the computational complexity (Qu & Hahn, 2009) associated with time-varying dynamics (e.g., fatigue) (Ding et al., 2003).

Thirdly, predictive simulations of FES-driven or FES-assisted motions (e.g., walking, cycling, reaching, and grasping) require the coupling of FES models with the equations of motion as well as adequate muscle force-length-velocity relationships. Predictive simulations are usually actuated through Hill-type muscle models (Wakeling et al., 2023). A package capable of replacing muscle actuation by FES models in multibody musculoskeletal models will allow us to simulate realistic FES-driven tasks.

Fourthly, personalized rehabilitation strategy is required to facilitate the motor recovery. Therefore, identifying the patient-specific muscle response to FES is a crucial step. Unfortunately, current complex identification methods are a barrier to clinical translation (Le et al., 2010). Providing a robust and customizable framework for the development of more patient-friendly protocols would help to overcome this barrier.

Despite its potential, optimal control-driven FES remains unadopted in clinical practice due to its low technology readiness level (Co et al., 2025). Cocofest is a comprehensive package designed to bridge the gaps and foster clinical adoption. It integrates nonlinear muscle dynamics dedicated to FES, manages muscle fatigue, interfaces FES with musculoskeletal models, supports customizable cost functions and parameter identification routines. With the goal of bringing this technology to patient care, we believe this package will contribute to the open-science effort. Cocofest is expected to accelerate the increase of technology readiness level by strengthening knowledge foundation.

State of the Field

Several open-source toolkits support optimal control computations for musculoskeletal biomechanics, such as: - OpenSim Moco (Dembia et al., 2020), a C++ OpenSim extension that enables motion tracking and prediction using efficient direct-collocation formulations coupled to nonlinear programming solvers. - SCONE (Geijtenbeek, 2019), a C++/C predictive-simulation environment for human and animal motion that optimizes neuromusculoskeletal controllers to achieve task-level objectives (e.g., stable walking at a target speed). - Bioptim (Michaud et

74 [al., 2022](#)), a Python optimal-control framework for biomechanics that supports both direct
75 collocation and multiple shooting, with flexible interfaces to nonlinear programming solvers.

76 However, these toolkits are not tailored for FES. They control muscle activation as a piecewise
77 linear/constant excitation, whereas FES requires optimizing deliverable stimulation patterns
78 under device and safety constraints. As a result, they lack reusable, validated components
79 for the stimulation-to-force pathway and fatigue/recovery dynamics, limiting reproducible
80 comparison of FES models and slowing translation to practical stimulation design. Cocofest
81 addresses this gap by implementing published FES models that can drive musculoskeletal
82 models. This design supports reproducible comparisons of FES modeling assumptions and
83 accelerates prototyping of patient- and task-specific stimulation optimization. Cocofest also
84 includes utilities for model identification and receding-horizon optimization to support FES
85 research workflows.

86 Software Design

87 Cocofest is a Python library that relies on Biorbd, a musculoskeletal physics engine ([Michaud
88 & Begon, 2021](#)), and Bioptim, an open-source optimization framework for biomechanical
89 problems ([Michaud et al., 2022](#)). Specifically, Bioptim enables easy OCP customization
90 including cost functions, bounds, constraints, transcription methods (e.g., direct collocation),
91 integration methods, and solving methods (e.g., full- and receding-horizon OCPs).

92 In conventional Hill-type muscle model, muscle force (F_m) is the product of a the muscle
93 activation, F_{max} the maximal isometric muscle force, f_l the force-length, f_v the force-velocity
94 and f_{pas} the passive force-length relationship: $F_m(t) = a(t) F_{max} f_l(\tilde{l}_m) f_v(\tilde{v}_m) + f_{pas}(\tilde{l}_m)$.
95 Cocofest replaces $a(t) \times F_{max}$ by the force obtained using FES models. This approach allows
96 motions driven-FES simulations, meanwhile benefiting from musculoskeletal model properties
97 (e.g., muscle insertion, inertial parameters).

98 Cocofest was developed to maintain a consistent structure between classes and functions to
99 facilitate the OCP customization and new FES model implementation. This shared interface
100 promotes reproducible work and comparisons of optimal control-driven FES strategies.

101 Research Impact Statement

102 Cocofest was developed to address several gaps in the literature, including the lack of
103 systematic comparisons of FES models and OCP formulations, accessible tools for FES model
104 identification, and open-source software for reproducible research. It enables researchers
105 to generate personalized stimulation patterns, compare alternative OCP formulations, and
106 simulate realistic FES-driven tasks. By providing a consistent software structure and clear
107 documentation, Cocofest aims to streamline research workflows and support translation toward
108 FES rehabilitation applications. Although the project is new and targets a niche domain, it
109 already offers a shared, reproducible environment that can foster discussion, collaboration, and
110 broader adoption of open-source practices within the FES community, which is an important
111 step toward clinical translation of this technique ([Co et al., 2025](#)).

112 AI Usage Disclosure

113 The authors used ChatGPT only to improve the manuscript clarity and readability. After
114 using this tool/service, the authors reviewed and edited the content as needed and took full
115 responsibility for the content of the publication.

116 GitHub Copilot and ChatGPT were used to assist in code refactoring and documentation.
117 Authors made all the core design and architectural decisions.

Acknowledgements

The package development was supported by the Fonds de recherche du Québec – Nature et technologies (FRQNT, Grant 341023) and by the FRQ strategic group in Ingénierie de technologies interactives en réadaptation (INTER #160 OptiStim).

References

- Co, K., Begon, M., Bailly, F., & Moissenet, F. (2025). Optimal control-driven functional electrical stimulation: A PRISMA-ScR scoping review. *Computers in Biology and Medicine*, 199, 111311. <https://doi.org/10.1016/j.combiomed.2025.111311>
- Dembia, C. L., Bianco, N. A., Falisse, A., Hicks, J. L., & Delp, S. L. (2020). OpenSim moco: Musculoskeletal optimal control. *PLOS Computational Biology*, 16(12), e1008493. <https://doi.org/10.1371/journal.pcbi.1008493>
- Ding, J., Wexler, A. S., & Binder-Macleod, S. A. (2003). Mathematical models for fatigue minimization during functional electrical stimulation. *Journal of Electromyography and Kinesiology*, 13(6), 575–588. [https://doi.org/10.1016/S1050-6411\(03\)00102-0](https://doi.org/10.1016/S1050-6411(03)00102-0)
- Geijtenbeek, T. (2019). SCONE: Open source software for predictive simulation of biological motion. *Journal of Open Source Software*, 4(38), 1421. <https://doi.org/10.21105/joss.01421>
- Hunt, K. J., Munih, M., & Donaldson, N. de N. (1997). Feedback control of unsupported standing in paraplegia. I. Optimal control approach. *IEEE Transactions on Rehabilitation Engineering*, 5(4), 331–340. <https://doi.org/10.1109/86.650287>
- Ibitoye, M. O., Hamzaid, N. A., Hasnan, N., Abdul Wahab, A. K., & Davis, G. M. (2016). Strategies for rapid muscle fatigue reduction during FES exercise in individuals with spinal cord injury: A systematic review. *PloS One*, 11(2), e0149024. <https://doi.org/10.1371/journal.pone.0149024>
- Le, F., Markovsky, I., Freeman, C. T., & Rogers, E. (2010). Identification of electrically stimulated muscle models of stroke patients. *Control Engineering Practice*, 18(4), 396–407. <https://doi.org/10.1016/j.conengprac.2009.12.007>
- Michaud, B., Bailly, F., Charbonneau, E., Ceglia, A., Sanchez, L., & Begon, M. (2022). Bioptim, a python framework for musculoskeletal optimal control in biomechanics. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 53(1), 321–332. <https://doi.org/10.1109/TSMC.2022.3183831>
- Michaud, B., & Begon, M. (2021). Biorbd: A c++, python and matlab library to analyze and simulate the human body biomechanics. *Journal of Open Source Software*, 6(57), 2562. <https://doi.org/10.21105/joss.02562>
- Puchaud, P., Bailly, F., & Begon, M. (2023). Direct multiple shooting and direct collocation perform similarly in biomechanical predictive simulations. *Computer Methods in Applied Mechanics and Engineering*, 414, 116162. <https://doi.org/10.1016/j.cma.2023.116162>
- Qu, C. C., & Hahn, J. (2009). Computation of arrival cost for moving horizon estimation via unscented kalman filtering. *Journal of Process Control*, 19(2), 358–363. <https://doi.org/10.1016/j.jprocont.2008.04.005>
- Veltink, P. H., Chizeck, H. J., Crago, P. E., & El-Bialy, A. (1992). Nonlinear joint angle control for artificially stimulated muscle. *IEEE Transactions on Biomedical Engineering*, 39(4), 368–380. <https://doi.org/10.1109/10.126609>
- Wakeling, J. M., Febrer-Nafría, M., & De Groote, F. (2023). A review of the efforts to develop muscle and musculoskeletal models for biomechanics in the last 50 years. *Journal*

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